

The graded order law collapses at ζ_3 :
 $G_\lambda(\zeta_3) = 0$ if and only if $\lambda \in \{(3, 1), (2, 1, 1)\}$

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Abstract

Let $G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)}$ be the graded branch-exponent generating function, where $s(T) = \sum_{i \in \text{Des}(T)} w_i$ with the parity weights $w_i = 2i - 1$ if $n - i$ is odd and $w_i = 0$ otherwise. At $q = -1$ this polynomial obeys a rich vanishing law, $\text{ord}_{q=-1} G_\lambda = \lfloor |2\text{-core}(\lambda)|/2 \rfloor$, unbounded along the staircase of 2-cores. We prove that at a primitive *cube* root of unity ζ_3 the law collapses completely:

$$G_\lambda(\zeta_3) = 0 \iff \lambda \in \{(3, 1), (2, 1, 1)\},$$

and at those two shapes the order is exactly 1. In particular every partition of positive 3-weight (every non-3-core, and indeed all but two 3-cores) satisfies $\text{ord}_{\zeta_3} G_\lambda = 0$. The proof turns the apparent root-of-unity cancellation into a question of Schur positivity. A master identity expresses the fiber value as $G_\lambda(\zeta_3) = \langle s_\lambda, h_1^e \prod_k (h_2 + e_2 \zeta_3^{2k-1}) \rangle$; grouping the factors by residue class mod 3 and using the single quadratic identity $\Theta := (h_2 + e_2 \zeta_3)(h_2 + e_2 \zeta_3^2) = s_{(4)} + 2s_{(2,2)} + s_{(1^4)}$ (manifestly Schur-positive) reduces the value, in all cases, to a nonnegative integer inner product $\langle s_\lambda, h_1^A \Theta^b \rangle$. A Pieri computation and the fact that Θ^b has *full* Schur support for $b \geq 3$ finish the proof, the only residue being a finite check up to $n = 17$.

1 Setup

Throughout $\lambda \vdash n$, $\text{SYT}(\lambda)$ is the set of standard Young tableaux of shape λ , and for $T \in \text{SYT}(\lambda)$,

$$\text{Des}(T) = \{i \in [n-1] : i+1 \text{ lies in a strictly lower row of } T \text{ than } i\}.$$

Fix the *active set* and the parity weights

$$A = \{k \in [n-1] : n-k \text{ odd}\}, \quad w_k = (2k-1) \mathbf{1}_{[k \in A]}, \quad s(T) = \sum_{i \in \text{Des}(T)} w_i,$$

and set $m = |A| = \lfloor n/2 \rfloor$, $e = n \bmod 2$. The object of study is

$$G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)} \in \mathbb{Z}_{\geq 0}[q].$$

We write $\zeta_3 = e^{2\pi i/3}$ for a primitive cube root of unity, $\Phi_3(q) = q^2 + q + 1$ for the third cyclotomic polynomial, and $\text{ord}_{\zeta_3} G_\lambda = \max\{m : \Phi_3^m \mid G_\lambda\}$ for the order of vanishing at ζ_3 .

We work in the ring Λ of symmetric functions, with Schur, complete, elementary and power-sum bases s_λ, h_r, e_r, p_r , and the Hall inner product $\langle \cdot, \cdot \rangle$, for which $\langle s_\lambda, s_\mu \rangle = \delta_{\lambda\mu}$, $\langle s_\lambda, h_\mu \rangle = K_{\lambda\mu}$ (Kostka numbers), and $\langle s_\lambda, p_\mu \rangle = \chi^\lambda(\mu)$. For symmetric functions f we write $\text{supp}(f) = \{\mu : \langle s_\mu, f \rangle \neq 0\}$; if all Schur coefficients of f are ≥ 0 we call f *Schur-positive*. We use freely $h_2 = \frac{1}{2}(p_1^2 + p_2)$, $e_2 = \frac{1}{2}(p_1^2 - p_2)$, so that $h_2 + e_2 = p_1^2 = h_1^2$ and $h_2 - e_2 = p_2$.

2 The master identity for the fiber value

The following is the value-at- ζ_3 specialization of the descent enumerator that drove the $q = -1$ law; the proof is identical in form, so we give it compactly.

Lemma 1 (Descent enumerator; Clio 2026-06-01). *For indeterminates $\eta = (\eta_1, \dots, \eta_{n-1})$, with $\alpha(R)$ the composition of n whose descent set is R ,*

$$\sum_{T \in \text{SYT}(\lambda)} \prod_{k \in \text{Des}(T)} \eta_k = \sum_{R \subseteq [n-1]} K_{\lambda, \alpha(R)} \prod_{k \in R} \eta_k \prod_{k \notin R} (1 - \eta_k).$$

Proposition 1 (Master identity). *For any complex number ζ ,*

$$G_\lambda(\zeta) = \left\langle s_\lambda, h_1^e \prod_{k \in A} (h_2 + e_2 \zeta^{2k-1}) \right\rangle.$$

Proof. Put $\eta_k = \zeta^{w_k}$ in Lemma 1. Then $\prod_{k \in \text{Des}(T)} \eta_k = \zeta^{\sum_{i \in \text{Des}(T)} w_i} = \zeta^{s(T)}$, so the left side is $G_\lambda(\zeta)$. For $k \notin A$ we have $w_k = 0$, hence $\eta_k = 1$ and the factor $1 - \eta_k$ vanishes; only $R \supseteq A^c$ survive. Write $R = A^c \sqcup U$ with $U \subseteq A$; for $k \in A$, $\eta_k = \zeta^{2k-1}$ and $1 - \eta_k = 1 - \zeta^{2k-1}$. As in the $q = -1$ proof, the content of $\alpha(A^c \cup U)$ is $(2^{m-|U|}, 1^{e+2|U|})$ (adjoining $k \in A$ to the marks of $\alpha(A^c)$ splits the 2-gap centred at k into two 1's), whence $K_{\lambda, \alpha(A^c \cup U)} = \left\langle s_\lambda, h_2^{m-|U|} h_1^{e+2|U|} \right\rangle$. Therefore

$$G_\lambda(\zeta) = \left\langle s_\lambda, h_1^e \sum_{U \subseteq A} h_2^{m-|U|} h_1^{2|U|} \prod_{k \in U} \zeta^{2k-1} \prod_{k \in A \setminus U} (1 - \zeta^{2k-1}) \right\rangle.$$

The sum over U factorises: each $k \in A$ contributes $h_1^2 \zeta^{2k-1}$ if $k \in U$ and $h_2(1 - \zeta^{2k-1})$ if $k \notin U$, so the bracket is $h_1^e \prod_{k \in A} (h_2(1 - \zeta^{2k-1}) + h_1^2 \zeta^{2k-1})$. Finally $h_2(1 - \zeta^{2k-1}) + h_1^2 \zeta^{2k-1} = h_2 + (h_1^2 - h_2) \zeta^{2k-1} = h_2 + e_2 \zeta^{2k-1}$. $\square$

At $\zeta = -1$ each factor is $h_2 - e_2 = p_2$ and one recovers $G_\lambda(-1) = \chi^\lambda(2^m 1^e)$; the cube-root case is governed instead by the following grouping.

3 Grouping by residue and the quadratic identity

For $\zeta = \zeta_3$ write $z_k = \zeta_3^{2k-1}$. Since $2k-1 \equiv 0, 1, 2 \pmod{3}$ according as $k \equiv 2, 1, 0 \pmod{3}$, the value z_k is $1, \zeta_3, \zeta_3^2$ respectively. Set

$$n_0 = \#\{k \in A : k \equiv 2\}, \quad n_1 = \#\{k \in A : k \equiv 1\}, \quad n_2 = \#\{k \in A : k \equiv 0\} \pmod{3},$$

so that $n_0 + n_1 + n_2 = m$, and define $\phi = h_2 + e_2 \zeta_3$, $\bar{\phi} = h_2 + e_2 \zeta_3^2$. Using $h_2 + e_2 = h_1^2$,

$$\prod_{k \in A} (h_2 + e_2 z_k) = (h_1^2)^{n_0} \phi^{n_1} \bar{\phi}^{n_2}. \tag{1}$$

Lemma 2 (The imbalance is at most one). $\delta := n_1 - n_2 \in \{0, 1\}$, with $\delta = 1$ if and only if $n \equiv 2 \pmod{3}$.

Proof. A is an arithmetic progression of common difference 2: it is $\{1, 3, \dots, n-1\}$ if n is even and $\{2, 4, \dots, n-1\}$ if n is odd. Replacing n by $n+6$ preserves the parity of n , hence the progression type, and appends to A exactly the three new terms $x, x+2, x+4$ of the correct parity lying in $[n, n+5]$. These three are consecutive terms of a step-2 progression, so modulo 3 (as $\gcd(2, 3) = 1$) they realise all three residues once each; thus each of n_0, n_1, n_2 increases by 1 and δ is unchanged. Hence δ depends only on $n \pmod{6}$, and a direct check of $n = 1, \dots, 6$ gives $\delta = (0, 1, 0, 0, 1, 0)$, i.e. $\delta = 1 \iff n \equiv 2, 5 \pmod{6} \iff n \equiv 2 \pmod{3}$. $\square$

Lemma 3 (Quadratic Schur-positivity). $\Theta := \phi \bar{\phi} = h_2^2 - h_2 e_2 + e_2^2 = \frac{p_1^4 + 3p_2^2}{4} = s_{(4)} + 2s_{(2,2)} + s_{(1^4)}$.

Proof. $\phi \bar{\phi} = h_2^2 + h_2 e_2 (\zeta_3 + \zeta_3^2) + e_2^2 = h_2^2 - h_2 e_2 + e_2^2$ since $\zeta_3 + \zeta_3^2 = -1$. Substituting $h_2 = \frac{1}{2}(p_1^2 + p_2)$, $e_2 = \frac{1}{2}(p_1^2 - p_2)$ gives $\frac{1}{4}(p_1^4 + 3p_2^2)$. For the Schur expansion use Jacobi–Trudi / Pieri: $h_2^2 = s_4 + s_{31} + s_{22}$, $e_2^2 = s_{22} + s_{211} + s_{1^4}$, $h_2 e_2 = s_{31} + s_{211}$; hence $h_2^2 - h_2 e_2 + e_2^2 = s_4 + 2s_{22} + s_{1^4}$. $\square$

4 Reduction to a Schur-support statement

Combining Proposition 1, (1) and Lemmas 2–3, set

$$a := e + 2n_0 \quad (\geq 0), \quad b := n_2 \quad (\geq 0).$$

Proposition 2 (Value as a Schur inner product). *With a, b as above:*

(A) *If $n \equiv 0, 1 \pmod{3}$ then $n_1 = n_2 = b$ and*

$$G_\lambda(\zeta_3) = \langle s_\lambda, h_1^a \Theta^b \rangle \in \mathbb{Z}_{\geq 0}.$$

(B) *If $n \equiv 2 \pmod{3}$ then $n_1 = n_2 + 1$, and writing $X = \langle s_\lambda, h_1^a \Theta^b h_2 \rangle$, $Y = \langle s_\lambda, h_1^a \Theta^b e_2 \rangle$ (both in $\mathbb{Z}_{\geq 0}$),*

$$G_\lambda(\zeta_3) = X + \zeta_3 Y.$$

Consequently $G_\lambda(\zeta_3) = 0$ if and only if $X = Y = 0$, and a sufficient condition for $G_\lambda(\zeta_3) \neq 0$ is

$$X + Y = \langle s_\lambda, h_1^{a+2} \Theta^b \rangle > 0.$$

Proof. By Lemma 2, in case (A) $\delta = 0$ so (1) equals $(h_1^2)^{n_0} \Theta^b$; multiplying by the prefactor h_1^e gives $h_1^a \Theta^b$, and pairing with s_λ gives the formula. The inner product is a nonnegative integer because $h_1^a \Theta^b$ is Schur-positive (a product of the Schur-positive functions h_1 and Θ , Lemma 3).

In case (B) $\delta = 1$, so (1) equals $(h_1^2)^{n_0} \Theta^b \phi = (h_1^2)^{n_0} \Theta^b (h_2 + \zeta_3 e_2)$; with the prefactor h_1^e this is $h_1^a \Theta^b (h_2 + \zeta_3 e_2)$, and pairing with s_λ yields $G_\lambda(\zeta_3) = X + \zeta_3 Y$ with X, Y as stated, again nonnegative integers. Since $\{1, \zeta_3\}$ is linearly independent over $\mathbb{Q}$ and $X, Y \in \mathbb{Z}$, we have $G_\lambda(\zeta_3) = 0 \iff X = Y = 0$. Finally $X + Y = \langle s_\lambda, h_1^a \Theta^b (h_2 + e_2) \rangle = \langle s_\lambda, h_1^{a+2} \Theta^b \rangle$ because $h_2 + e_2 = h_1^2$; as $X, Y \geq 0$, $X + Y > 0$ forces $(X, Y) \neq (0, 0)$, hence $G_\lambda(\zeta_3) \neq 0$. $\square$

Both cases thus rest on the single quantity $\langle s_\lambda, h_1^A \Theta^b \rangle$ with $A + 4b = n$ (here $A = a$ in case (A) and $A = a + 2$ in case (B)). We characterise its positivity.

Lemma 4 (Pieri support criterion). *For $A \geq 0$,*

$$\langle s_\lambda, h_1^A \Theta^b \rangle = \sum_{\substack{\mu \subseteq \lambda \\ |\mu| = 4b}} \langle s_\mu, \Theta^b \rangle f^{\lambda/\mu},$$

where $f^{\lambda/\mu} = \# \text{SYT}(\lambda/\mu)$. Hence $\langle s_\lambda, h_1^A \Theta^b \rangle > 0$ if and only if λ contains some $\mu \in \text{supp}(\Theta^b)$ with $|\mu| = 4b$.

Proof. Since $h_1^A = p_1^A$ and $s_\mu p_1^A = \sum_\lambda f^{\lambda/\mu} s_\lambda$ (the number of standard skew tableaux of shape λ/μ), writing $\Theta^b = \sum_\mu \langle s_\mu, \Theta^b \rangle s_\mu$ and pairing with s_λ gives the displayed sum. All $\langle s_\mu, \Theta^b \rangle \geq 0$ (Schur positivity) and $f^{\lambda/\mu} > 0 \iff \mu \subseteq \lambda$. So the sum is positive iff some term is, i.e. iff some $\mu \in \text{supp}(\Theta^b)$ of size $4b$ is contained in λ . $\square$

5 Full support of Θ^b for $b \geq 3$

Lemma 5 (Removable strips). *A partition ρ admits a removable horizontal strip of size 4 (i.e. some $\mu \subseteq \rho$ with ρ/μ a horizontal 4-strip) if and only if $\rho_1 \geq 4$, and a removable vertical 4-strip if and only if $\rho'_1 \geq 4$. Consequently, if $|\rho| > 9$ then ρ admits a removable horizontal or vertical 4-strip.*

Proof. Taking $\mu = (\rho_2, \rho_3, \dots)$ makes ρ/μ a horizontal strip of size $\sum_i (\rho_i - \rho_{i+1}) = \rho_1$, and any intermediate $\mu_i \in [\rho_{i+1}, \rho_i]$ keeps μ a partition with ρ/μ horizontal; thus removable horizontal strips of every size $0, 1, \dots, \rho_1$ exist, and none larger. The vertical statement is the conjugate. If $\rho_1 \leq 3$ and $\rho'_1 \leq 3$ then $\rho \subseteq (3, 3, 3)$, so $|\rho| \leq 9$; contrapositively $|\rho| > 9$ forces $\rho_1 \geq 4$ or $\rho'_1 \geq 4$. $\square$

Lemma 6 (Full support). *For every $b \geq 3$, $\text{supp}(\Theta^b)$ is all of $\{\mu : \mu \vdash 4b\}$.*

Proof. Base $b = 3$. Expanding Θ^3 in the Schur basis (a finite Pieri/LR computation; all 77 partitions of 12 occur, see §7) shows $\text{supp}(\Theta^3)$ is full. *Induction.* Assume $\text{supp}(\Theta^b)$ is full, $b \geq 3$, and let $\rho \vdash 4(b+1)$. Then $|\rho| = 4b + 4 \geq 16 > 9$, so by Lemma 5 ρ has a removable horizontal or vertical 4-strip; let $\mu \subseteq \rho$, $\mu \vdash 4b$, be the corresponding partition, so that ρ/μ is a horizontal (resp. vertical) 4-strip and hence the Littlewood–Richardson coefficient $c_{\mu, (4)}^\rho = 1$ (resp. $c_{\mu, (1^4)}^\rho = 1$) by Pieri. Since (4) and (1^4) lie in $\text{supp}(\Theta)$ with positive coefficient and $\mu \in \text{supp}(\Theta^b)$ by hypothesis,

$$\langle s_\rho, \Theta^{b+1} \rangle = \sum_{\mu', \nu} \langle s_{\mu'}, \Theta^b \rangle \langle s_\nu, \Theta \rangle c_{\mu', \nu}^\rho \geq \langle s_\mu, \Theta^b \rangle \langle s_{(4) \text{ or } (1^4)}, \Theta \rangle c_{\mu, \cdot}^\rho > 0,$$

all summands being nonnegative. Thus $\rho \in \text{supp}(\Theta^{b+1})$, and $\text{supp}(\Theta^{b+1})$ is full. $\square$

Remark 1. The hypothesis $b \geq 3$ is sharp: $\text{supp}(\Theta^2)$ omits exactly $(6, 1, 1)$, $(4, 2, 1, 1)$, $(3, 3, 2)$, $(3, 1^6)$, and $\text{supp}(\Theta) = \{(4), (2, 2), (1^4)\}$ omits $(3, 1)$, $(2, 1, 1)$. This is the source of the two exceptional shapes below.

6 Main theorem

Theorem 1. *For every partition λ ,*

$$G_\lambda(\zeta_3) = 0 \iff \lambda \in \{(3, 1), (2, 1, 1)\}.$$

At these two shapes $G_\lambda(q) = q^5 + q + 1 = \Phi_3(q)(q^3 - q^2 + 1)$ and $\text{ord}_{\zeta_3} G_\lambda = 1$. For every other λ , $\text{ord}_{\zeta_3} G_\lambda = 0$. In particular, if λ has positive 3-weight (i.e. λ is not a 3-core) then $\text{ord}_{\zeta_3} G_\lambda = 0$.

Proof. The set $\{n : b = n_2 \leq 2\} = \{1, \dots, 15\} \cup \{17\}$ is finite (it has maximum 17; note $n = 16$ already has $n_2 = 3$); for every other n one has $b \geq 3$.

Case $b \geq 3$ (i.e. $n \notin \{1, \dots, 15, 17\}$). Let $\lambda \vdash n$ be arbitrary. By Proposition 2 the nonvanishing of $G_\lambda(\zeta_3)$ follows (in case (A) directly, in case (B) via $X+Y$) from $\langle s_\lambda, h_1^A \Theta^b \rangle > 0$ with $A = n - 4b \geq 0$. By Lemma 6, $\text{supp}(\Theta^b)$ is full, so every $\mu \vdash 4b$ lies in it; and λ , having $n = A + 4b \geq 4b$ boxes, contains some $\mu \subseteq \lambda$ with $|\mu| = 4b$ (remove A removable corner boxes one at a time). By Lemma 4, $\langle s_\lambda, h_1^A \Theta^b \rangle > 0$. Hence $G_\lambda(\zeta_3) \neq 0$.

Case $b \leq 2$ (i.e. $n \in \{1, \dots, 15, 17\}$). This is a finite computation (§7): over all $\lambda \vdash n$ with $n \leq 17$, the only shapes with $G_\lambda(\zeta_3) = 0$ are $(3, 1)$ and $(2, 1, 1)$, both 3-cores. Concretely, for the four shapes of $n = 4$, $\langle s_\lambda, \Theta \rangle$ vanishes only for $\lambda \in \{(3, 1), (2, 1, 1)\}$ (Lemma 3), and a check of cases (A)/(B) for the remaining $n \in \{1, \dots, 15, 17\}$ exhibits a containing $\mu \in \text{supp}(\Theta^b)$ for every λ .

Combining, $G_\lambda(\zeta_3) = 0$ iff $\lambda \in \{(3, 1), (2, 1, 1)\}$. (At $n = 4$, in case (A), $\langle s_\lambda, \Theta \rangle$ is the coefficient of s_λ in $\Theta = s_{(4)} + 2s_{(2,2)} + s_{(1^4)}$, which is zero precisely for the two shapes $(3, 1), (2, 1, 1)$ omitted from $\text{supp}(\Theta)$.) The orders are read off from the explicit polynomials. At $\lambda = (3, 1)$ the three tableaux have descent sets $\{1\}, \{2\}, \{3\}$ and weights $(w_1, w_2, w_3) = (1, 0, 5)$, so

$$G_{(3,1)}(q) = q^5 + q + 1 = \Phi_3(q)(q^3 - q^2 + 1), \quad G_{(2,1,1)}(q) = q^6 + q^5 + q = q\Phi_3(q)(q^3 - q + 1),$$

and in each case Φ_3 does not divide the displayed cofactor (its value at ζ_3 is $2 - \zeta_3^2 \neq 0$, resp. $2 - \zeta_3 \neq 0$), so $\text{ord}_{\zeta_3} G_{(3,1)} = \text{ord}_{\zeta_3} G_{(2,1,1)} = 1$. For all other λ , $G_\lambda(\zeta_3) \neq 0$ gives $\text{ord}_{\zeta_3} G_\lambda = 0$. Since both exceptional shapes are 3-cores, every partition of positive 3-weight has $\text{ord}_{\zeta_3} G_\lambda = 0$. $\square$

7 Computational verification

All steps were checked by exact computation; scripts in `~/projects/scratch/(prove-master-id-check.py, prove-reformulation-check.py, prove-schur-support.py)`.

- **Master identity (Prop. 1).** The character form $G_\lambda(\zeta_3) = \sum_{B \subseteq A} \chi^\lambda(2^m - |B| 1^{2|B|+e}) \prod_{k \in B} \frac{1+z_k}{2} \prod_{k \notin B} \frac{1-z_k}{2}$ agrees with direct SYT evaluation of $G_\lambda(\zeta_3)$ for all $\lambda \vdash n$, $n \leq 8$ (66 shapes, 0 mismatches).
- **Reformulation (Prop. 2).** For all $\lambda \vdash n$, $n \leq 11$ (194 shapes), the case-(A)/(B) values equal $G_\lambda(\zeta_3)$ from direct enumeration, and each of the quantities G (case A), X, Y (case B) is a nonnegative integer; 0 exceptions.
- **Lemma 2.** $\delta \in \{0, 1\}$ with $\delta = 1 \iff n \equiv 2 \pmod{3}$ for all $n \leq 200$; and $\{n : n_2 \leq 2\} = \{1, \dots, 15, 17\}$.
- **Lemmas 5, 6.** Maximal removable horizontal strip of ρ equals ρ_1 (all ρ , $n \leq 10$); $\text{supp}(\Theta^3)$ and $\text{supp}(\Theta^4)$ are full; $\text{supp}(\Theta^2)$ omits exactly the four listed shapes.
- **Finite check (Theorem 1).** Over all $\lambda \vdash n$, $n \leq 17$ (case A via $\text{supp}(h_1^a \Theta^b)$ for $n \leq 16$; case B via (X, Y) for $n \in \{2, 5, 8, 11, 14, 17\}$), the only λ with $G_\lambda(\zeta_3) = 0$ are $(3, 1), (2, 1, 1)$. This is consistent with the independent exhaustive enumeration over $|\lambda| \leq 12$ recorded in the order-law program.

8 Remark: the obstruction for general $d \geq 4$

The master identity (Prop. 1) holds at any ζ_d , and the same grouping pairs conjugate factors into

$$(h_2 + e_2 \omega^j)(h_2 + e_2 \omega^{-j}) = h_2^2 + c h_2 e_2 + e_2^2 = s_{(4)} + (1+c) s_{(3,1)} + 2 s_{(2,2)} + (1+c) s_{(2,1,1)} + s_{(1^4)},$$

with $\omega = \zeta_d$ and $c = 2 \cos(2\pi j/d)$. This quadratic factor is Schur-positive precisely when $c \geq -1$, i.e. $|j|/d \leq 1/3$. For $d = 3$ the only paired frequency is $j = 1$, giving $c = -1$ exactly; the off-diagonal coefficient $1+c$ vanishes and one lands on the clean $\Theta = s_{(4)} + 2s_{(2,2)} + s_{(1^4)}$ that makes the whole argument run. For $d \geq 4$ some paired frequencies have $c < -1$, so the individual factors need not be Schur-positive, and the reduction to a single nonnegative inner product breaks down. The collapse $\text{ord}_{\zeta_d} G_\lambda \in \{0, 1\}$ is verified computationally for all $d \leq 8$ and $|\lambda| \leq 12$, but a proof for general d requires controlling the (possibly signed) contributions of these factors; we leave this open.